

Sprint Report 3

PenTestPocket

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October 12th 2025 - November 7th 2025

Section 1: What's New (User Facing)

1.1 Feature - Nmap and Ncat

One of the many new additions to PTP was Nmap and Ncat, these tools were the first to be implemented The interface for PTP, and these tools have been connected using QSubprocess which will be explained in the developer section

1.2 Feature - curl and hashcat

The next tools added to PTP, that we additions near the end of Sprint 3, are curl and hashcat. While these tools are integrated and can be sent targets and run like expected within PTP, they need more customization options that will be added in Sprint 4

1.3 Feature - Data Conversions

One of the largest additions in Sprint 3 were all of the data conversion capabilities that were added; What this means is that I have added ways to convert one format to another, I.E. Text to Binary, or Binary to Hex, or so on and so forth. While there are still some kinks to work out, majority of the functionality is fully developed

1.4 Feature - Hashing

Hashing has been added with six different ways to hash a string, SHA1, SHA256, MD5, CRC-32, RIPEMD-160, and Shake256 (which replaced the original B-crypt). B-crypt was replaced as I wanted to minimize the amount of external downloads needed to run PTP, currently requiring only 4 (nmap, ncat, hashcat, curl)

Section 2: Work Summary (Developer Facing)

In **Sprint 3**, I integrated the tools and functions as mentioned in *#Section 1* using QProcess and QThread to create subprocesses for tools like nmap and threads for function calls like hashing respectively.

This was all accomplished starting from August 21st 2025, time tracking is stored in GitHub to break down how much time was spent in each grouping (specified in the roadmap) and how much each day.

Section 3: Unfinished Work

In **Sprint 3** I was not able to finish user logging, tool run data, pinned tools, and the final few tools to add to meet the minimum planned completion threshold

The default reason, which is true for these issues as well, is that for any delayed issue it is due to the tasks taking longer than expected; any other arising issues will be mentioned within the comments of the issue itself. For this sprint, a lot of delays came in the form of goal changes, other coursework, and personal commitments

NOTE - There are open issues, but they are meant to be placed into the final sprint, so they are not in scope for this sprint.

Section 4: Completed Issues/User Stories

Here are links to the issues that I completed in this sprint:

4.1 Issue links

Issue #18 - Weekly Coaching Session - 10/16

Issue #19 - Weekly Coaching Session - 10/23

Issue #20 - Weekly Coaching Session 10/30

Issue #48 - Assign Nmap Commands

Issue #52 - Integrate Ncat

Issue #57 - Add data conversion

Issue #60 - Integrate curl

Issue #64 - Add hashing

Issue #65 - Integrate Hashcat

Section 5: Incomplete Issues/User Stories

Here are links to issues I worked on but did not complete in this sprint:

5.1 Issue links & Explanations

[Will be Completed] Issue #21 - Weekly Coaching Session - 11/06

Explanation: This is a meeting, this will be completed before end of sprint 3

[Will be Completed] Issue #31 - Sprint Three Report Due

Explanation: This is the Sprint Report Three, if you are seeing this, it is completed

[Will be Completed] Issue #36 - Project Draft Three Report Due

Explanation: Project Two Report will be submitted before November 7th (end of Sprint 3)

[Incomplete] Issue #53 - Assign Ncat Commands

Explanation: I was prioritizing integration of multiple tools

[Incomplete] Issue #55 - Create more in depth user logging

Explanation: Converted into stretch goal to focus on tool creation

[Incomplete] Issue #56 - Store tool output in a file for later

Explanation: I was prioritizing integration of multiple tools before output collection

[Incomplete] Issue #61 - Assign curl commands

Explanation: I was prioritizing integration of multiple tools before output collection

[Will be Completed] Issue #67 - Blocked time for sprint 3 and report draft 3

Explanation: Time reserved to work on the sprint and project report, will be completed as soon as both of those are finished, which will be in Sprint 3

Section 6: Code Files for Review

Please review the following code files, which were actively developed during this sprint, for quality:

Main PenTestPocket Python Code

Change Log Python Code

User Logging Python Code

Help Window Python Code

Service Scanning Python Code

Password Cracking Python Code

Web Enumeration Python Code

Encode/Decode Python Code

Create Hash Python Code

Section 7: Retrospective Summary

7.1 What went well

Implementation of hashing and data conversion - I was able to fully implement and integrate all data conversions and hashing functionality with only the minor change of grouping ASCII and UTF-8 into 'Text'

Generic code structure for scalability and dynamic capabilities - The structure of the code I had spent a lot of sprint 1 and 2 setting up has allowed me to make changes, add new functions, and more with great ease and minimal update time

Additions of new tools - I was able to add a great number of tools and capabilities, roughly 10-15, compared to the previous sprint where I only had a single non-functional tool

7.2 What needs improvement

User logging - All actions by made within an instance of PTP need to be tracked and stored within a txt file that is viewable by the user at all times

Tool pinning - The main PTP window has four boxes reserved for pinned tools to be located for ease of use and access, this however only displays the name of the text and has no functionality

Tool history - Tools need to have the output stored for future use even after the current instance has ended so that it can be viewed later

7.3 Planned changes leading into Final Sprint 4

Final Tool Integrations - The remaining tools need to be implemented so that all non-tool related features can be implemented as early as possible as we are reaching the deadline for the capstone project

History, Logs, and Pins - After all tools are integrated, I will then need to add the three things mentioned above in *What needs improvement*

Stretch Goals - At this stage all the required goals will be implemented and all that will be needed after that is going to be stretch goals such as notepad, terminal, etc.